

3.6.2016(E)

K. J. Somaiya College of Engineering, Mumbai-77
(Autonomous College Affiliated to University of Mumbai)

Max. Marks: 100

Duration: 3 Hrs

End Semester Exam Partial Auto KT
April-May(2016)

Class: SE

Semester: III

Name of the Course: Applied Mathematics III

Branch: Comp & IT

Course Code: UCEC301/ UITC301

MU C B O S

Instructions:

- (1) All Questions are Compulsory
(2) Figures to right indicate full marks. Each sub-question has equal marks.

Question No.		Max. Marks
Q.1	Attempt any THREE of the following	21
(a)	Find Laplace transform of (i) $f(t) = te^{-4t} \sin 3t$ (ii) $f(t) = \frac{1}{t}(e^{-at} - e^{-bt})$	
(b)	Evaluate $L^{-1}\{2 \tanh^{-1} s\}$	
(c)	Solve using Laplace Transform : $(D^2 - 3D + 2)y = 4e^{2t}$ where $y(0) = -3, y'(0) = 5$	
(d)	Find inverse Laplace transform $\frac{s^2+2s+3}{(s^2+2s+2)(s^2+2s+5)}$	
(e)	Define periodic function and find Laplace transform of $f(t) = \begin{cases} 1; & 0 \leq t < a \\ -1; & a < t < 2a \end{cases}$ if $f(t)$ is periodic function with period $2a$.	
Q.2	Attempt any THREE of the following	21
(a)	Find Fourier Series of $f(x) = x^2$ in $(0, 2\pi)$ and Hence deduce that $\frac{\pi^2}{12} = \frac{1}{1^2} - \frac{1}{2^2} + \frac{1}{3^2} - \frac{1}{4^2} + \dots$	
(b)	Obtain Complex form of Fourier Series for $f(x) = e^{ax}$ in $(-\pi, \pi)$ where a is not an integer.	
(c)	Obtain half range sine series for $f(x) = x(\pi - x)$ in $(0, \pi)$ and hence show that $\frac{1}{1^6} + \frac{1}{2^6} + \frac{1}{3^6} + \dots = \frac{\pi^6}{945}$	
(d)	Obtain Fourier Series for $f(x) = \begin{cases} a(x-l), & -l < x < 0 \\ a(x+l), & 0 < x < l \end{cases}$ Hence deduce that $\frac{\pi}{4} = 1 - \frac{1}{3} + \frac{1}{5} - \frac{1}{7} + \dots$	

	(e) Is the set of functions $\{\sin x, \sin(2x), \sin(3x) \dots \dots \dots\}$ is orthogonal over $[-\pi, \pi]$?. If yes, construct Orthonormal set of functions.	
Q.3	Attempt any THREE of the following	21
	(a) Show that $3x^2y - y^3$ is harmonic and also find its harmonic conjugate.	
	(b) Find the orthogonal trajectory of the family of curves $x^3y - xy^3 = c$	
	(c) State the condition for a function to be analytic and hence find the constants a, b, c, d if $f(z) = x^2 + 2axy + by^2 + i(cx^2 + 2dxy + y^2)$ is an analytic function.	
	(d) Prove that $w = i \frac{(1-z)}{(z+1)}$ maps the interior of the unit circle in Z-plane onto the upper half of the W-plane.	
	(e) Find bilinear transformation which maps $z = \infty, i, 0$ on to $w = 0, i, \infty$. Also find its fixed points.	
Q.4	Attempt any THREE of the following	21
	(a) Define irrotational field and hence check whether the vector field $\vec{F} = (x + 2y + 4z)\vec{i} + (2x - 3y - z)\vec{j} + (4x - y + 2z)\vec{k}$ is irrotational.	
	(b) Evaluate by Green Theorem $\int_C \vec{F} \cdot d\vec{r}$ where C is the closed curve of the region bounded by $y = 4$ & $y = x^2$ in the xy plane and $\vec{F} = (x^2 - y)\vec{i} + (2y^2 + x)\vec{j}$	
	(c) Using Stokes theorem evaluate $\int_C \vec{F} \cdot d\vec{r}$ Where C is the boundary of the rectangle $x = 0, y = 0, x = a, y = b$ where $\vec{F} = x^2\vec{i} + xy\vec{j}$.	
	(d) Define coplanar vectors and prove that $\vec{a} \times \vec{b}, \vec{b} \times \vec{c}$ and $\vec{c} \times \vec{a}$ are coplanar vectors if $\vec{a}, \vec{b}, \vec{c}$ are coplanar.	
	(e) If $\vec{a}$ is constant vector, prove that $\nabla \times [\vec{r} \times (\vec{a} \times \vec{r})] = 3\vec{r} \times \vec{a}$	
Q.5	Attempt any Two of the following	16
	(a) Find the Z- transform of (i) $f(k) = k\alpha^k, k \geq 0$ (ii) $f(k) = {}^nC_k, k \geq 0$	
	(b) Find the inverse Z- transform of $f(z) = \frac{2z^2 - 10z + 13}{(z-3)^2(z-2)}$ If R.O.C. is (i) $ z < 2$ (ii) $2 < z < 3$ (iii) $ z > 3$ 6	
	(c) Prove that $Z(\cos bk) = \frac{z(z - \cos b)}{z^2 - 2z \cos b + 1}$ and hence find $Z(e^{-ak} \cos bk)$	